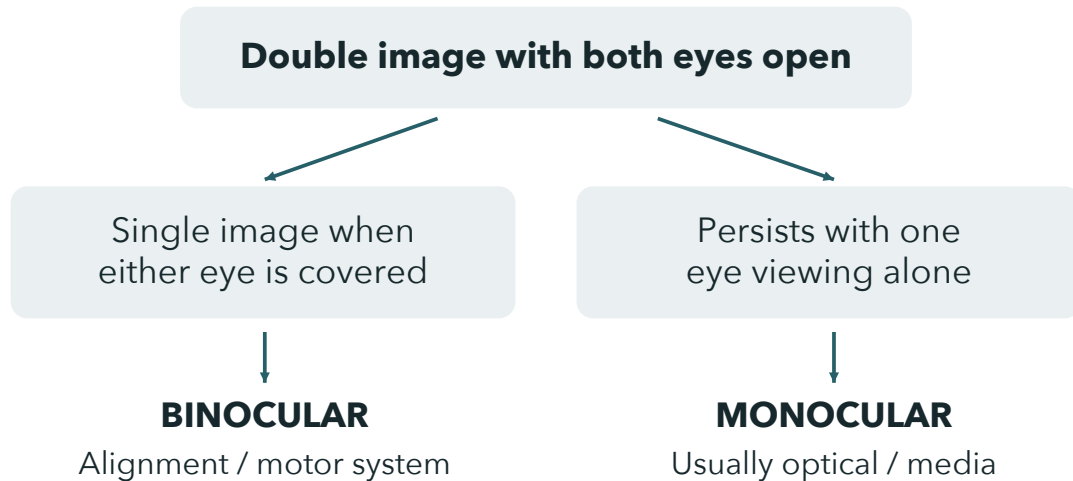


Neuro ophthalmology and orbit

CLINICALS | Localise the deficit, identify the threat, choose the next investigation.

Diplopia: start with one eye covered

One distinction changes the pathway



Original schematic. Cover each eye in turn to separate binocular from monocular diplopia.

Pattern → localisation

Finding	Most useful localisation
Ptosis; weak adduction, elevation and depression	CN III. Record pupil size/reactivity, completeness and associated neurological signs.
Horizontal diplopia worse at distance and towards the weak abducting eye	CN VI; assess for raised ICP, other neurological signs and orbital restriction.
Vertical/torsional diplopia worse on looking down in adduction	CN IV. Ask about trauma and compensatory head tilt.
Impaired adduction with abducting nystagmus in the other eye	Internuclear ophthalmoplegia: medial longitudinal fasciculus; brainstem disease.
Variable ptosis/diplopia, fatigability, pupils unaffected	Ocular myasthenia; ask about bulbar symptoms and breathing difficulty.

Focused examination

Test acuity, pupils, lids, globe position, alignment and ductions. Pain, headache or focal neurological deficits raise urgency. Proptosis with restriction suggests orbital disease.

References NANOS: Double vision | NANOS: Myasthenia gravis | Patel and Burdon: Modern assessment of isolated III palsy

Acute neuro signs: do not wait for recovery

Time-critical decisions

- 1 Acute acquired III palsy → urgent specialist assessment and CTA/MRA to exclude aneurysm; partial palsy or pupil involvement is especially concerning.
- 2 Acute painful Horner syndrome → emergency head/neck vascular imaging for carotid dissection. Do not wait for pupil-drop confirmation.
- 3 New homonymous field loss or diplopia with focal neurological deficits → acute stroke pathway.
- 4 Sudden headache, visual loss, ophthalmoplegia and possible hypotension → consider pituitary apoplexy; urgent imaging and endocrine/neurosurgical assessment.

Separate the clues

Pattern	Clinical implication
Poorly reactive large pupil + III weakness	Compression is concerning; normal pupils cannot exclude it.
Small pupil + mild ptosis; worse in dark	Horner syndrome; new head/neck pain suggests dissection.
Temporal loss in each eye	Chiasmal disease; investigate the sellar region.
Same field side lost in both eyes	Retrochiasmal disease; sudden onset can be stroke.

A useful caveat

Pupil sparing may support microvascular disease, but cannot safely exclude aneurysm, especially in partial palsy. Modern recommendations favour imaging acquired isolated III palsies at presentation. Isolated IV/VI palsy pathways depend on age and context.

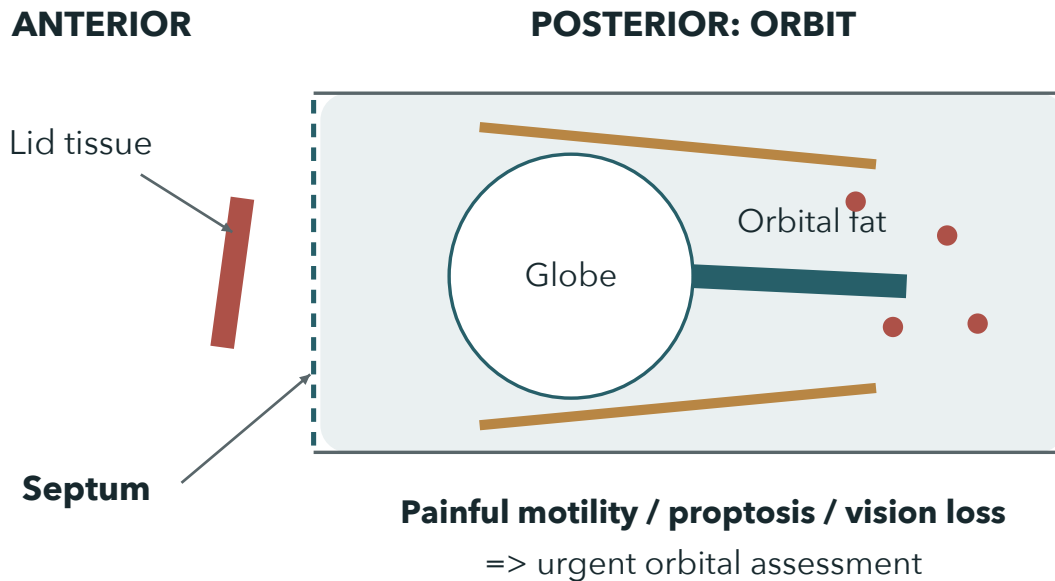
Field meridians

Chiasmal/retrochiasmal defects often respect the vertical meridian; many retinal nerve-fibre/optic-nerve defects follow the horizontal meridian. Interpret the whole pattern.

References Patel and Burdon: Modern assessment of isolated III palsy | AAO EyeWiki: Horner syndrome and urgent imaging | NANOS: Homonymous hemianopsia | NANOS: Pituitary tumour and apoplexy

Orbit: vision and motility determine urgency

The septum separates two management paths



Original orbital schematic, not to scale. A swollen lid alone cannot establish that infection is confined anteriorly.

Preseptal or orbital?

Finding	Meaning / next step
Normal vision; full painless movements; no proptosis	Supports preseptal cellulitis. A well patient may have oral antibiotics with reliable early review.
Painful/restricted movements, proptosis, diplopia, reduced acuity or RAPD	Orbital cellulitis: admit, IV antibiotics, urgent ophthalmology/ENT and contrast CT orbits/sinuses; brain imaging as indicated.
Cannot examine eye; or worsening on therapy	Orbital disease remains unexcluded: escalate urgently.
Bilateral progression, multiple motor palsies, V1/V2 loss or systemic toxicity	Consider cavernous sinus involvement; urgent specialist brain/venous imaging.

Thyroid eye disease can also threaten sight

Lid retraction, proptosis and restrictive diplopia suggest thyroid eye disease. Reduced acuity/colour/fields, RAPD or corneal breakdown need urgent assessment for optic neuropathy or exposure. Normal thyroid tests do not exclude eye disease.

Optic nerve: colour, pupils and fields

Four high-yield patterns

Pattern	Investigation / action
Subacute loss with painful eye movement, dyschromatopsia and RAPD; disc may look normal	Optic neuritis: prompt specialist assessment; MRI brain/orbits evaluates demyelination and alternative causes. Severe bilateral, recurrent or otherwise atypical disease needs a broader work-up.
Abrupt painless loss, often on waking, with an altitudinal defect and swollen disc	NAION is possible, but promptly assess for GCA in an older patient. Address vascular risks and sleep apnoea; there is no established routine treatment that reliably restores lost vision.
Severe sudden loss with pallid disc swelling, headache, jaw claudication or scalp tenderness	Suspect arteritic AION from GCA. Immediate systemic glucocorticoids and urgent ophthalmology/rheumatology care; do not wait for confirmatory testing.
Progressive loss, disproportionate colour deficit, pallor, proptosis or unusual fields	Consider compression. MRI brain/orbits with contrast, targeted by localisation.

When GCA is suspected

- 1 Draw ESR, CRP and full blood count/platelets promptly; do not delay treatment for blood sampling.
- 2 Start immediate systemic glucocorticoids via the acute pathway; visual loss may require IV treatment.
- 3 Arrange vascular ultrasound and/or temporal artery biopsy. Protect the fellow eye: treatment must not await confirmation.

Avoid two shortcuts

- An RAPD indicates asymmetric afferent dysfunction, not necessarily an abnormal-looking disc; dense asymmetric retinal disease can also cause it.
- Preserved central chart acuity does not exclude a clinically important optic-nerve field defect. Document colour and fields, not acuity alone.

Papilloedema: exclude secondary causes first

Paired swollen optic discs



Supplied M3 Anki note 1701959947007: A right eye, B left eye. Swollen discs require clinical confirmation of the cause. Supplied M3 Anki, media 95.

The safe diagnostic sequence

- 1 Confirm disc swelling and record BP, acuity, pupils, formal fields and disc appearance; assess urgently for neurological abnormalities.
- 2 Urgent MRI brain (or CT if MRI cannot be obtained promptly) PLUS MR/CT venography to exclude mass, hydrocephalus and cerebral venous sinus thrombosis.
- 3 After imaging excludes a contraindication, perform LP with lateral-decubitus opening pressure and CSF analysis. An elevated pressure alone does not establish IIH.

IIH is a diagnosis of exclusion

Headache, pulsatile tinnitus, transient visual obscurations and an isolated VI palsy fit raised ICP. Obesity or an empty sella does not prove IIH. Review medicines and secondary causes. Confirmed IIH care combines weight management, specialist pressure-lowering treatment and visual-field monitoring.

Protect vision

Worsening fields or rapidly declining vision require urgent specialist escalation; a normal central acuity reading is insufficient reassurance. A small crowded disc or buried drusen may mimic swelling, but uncertainty needs specialist assessment rather than delayed investigation.